

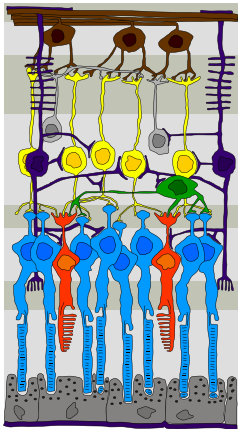
Yuli Multi-Enhancing Artificial Vision by Integrating Machine Learning in Retinal Prosthetics

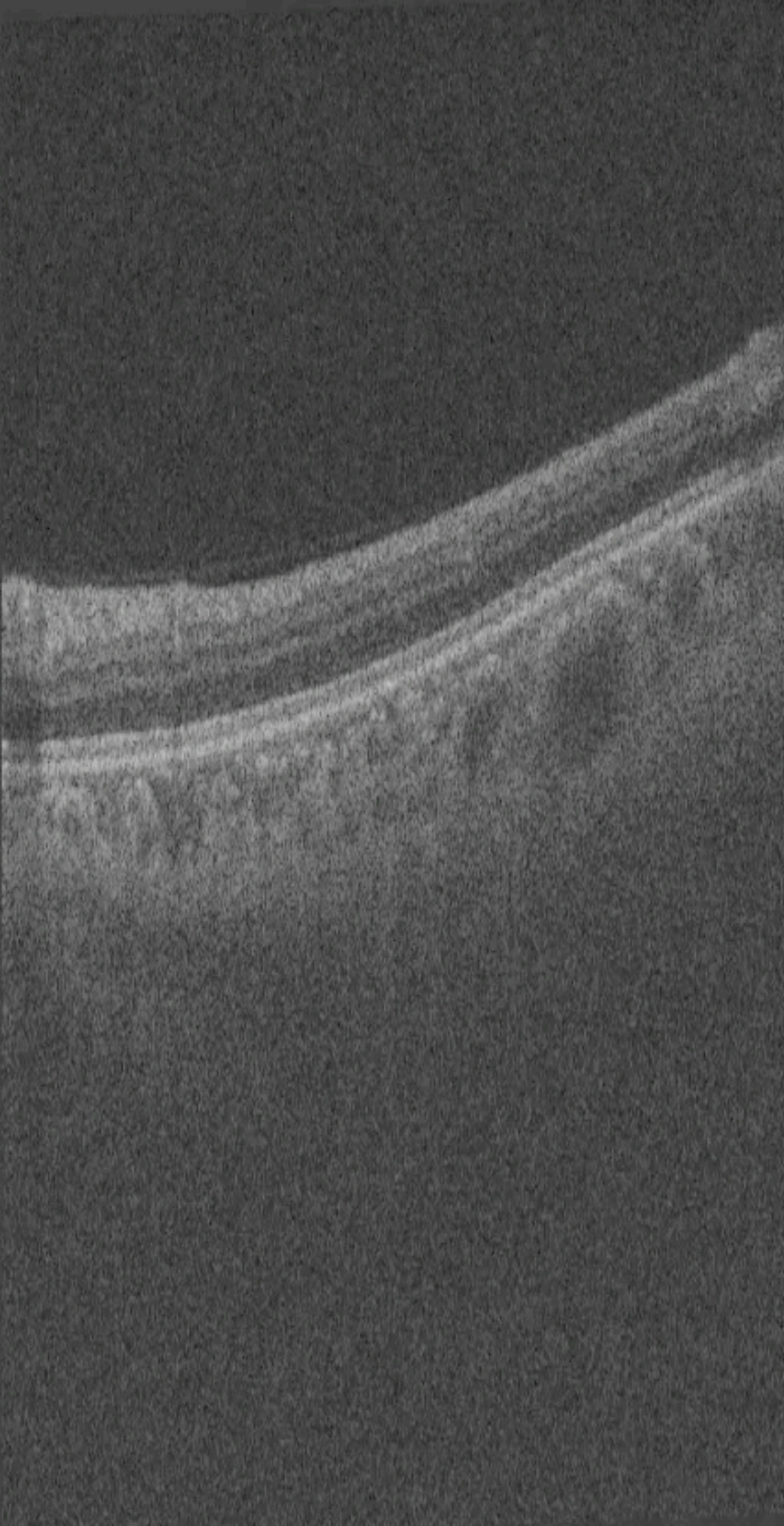
RWTHAACHEN
UNIVERSITY

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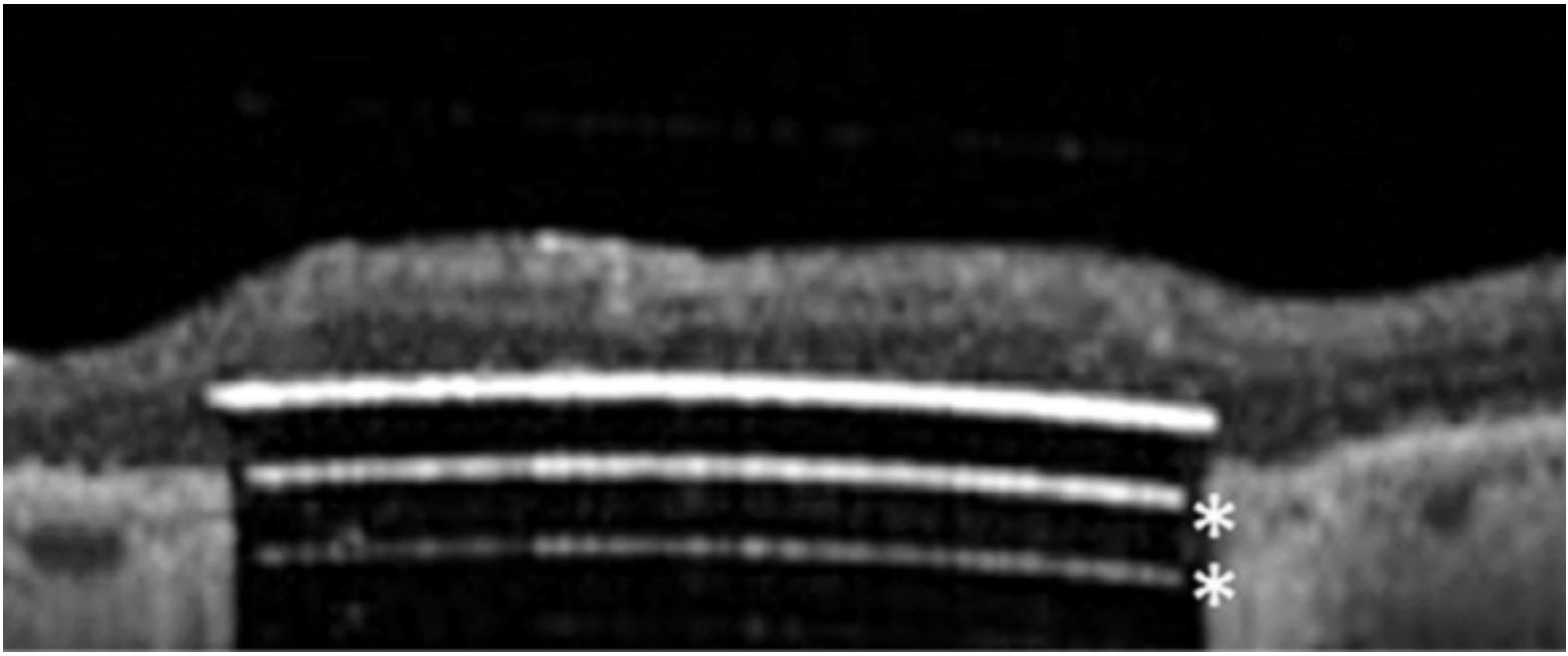
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Optical Coherence Tomography (OCT)



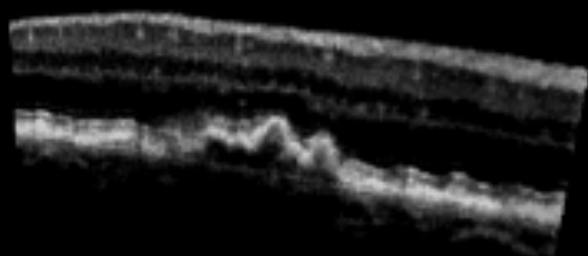


[Fang, et al., 2023]



[Mudpitai., 2024]

- Non-invasive, near-infrared light, 3D
- High resolution ($<10\text{ }\mu\text{m}$ axially, $<20\text{ }\mu\text{m}$ laterally)
- Important pre- and post-surgery
- Shared machine learning methodology

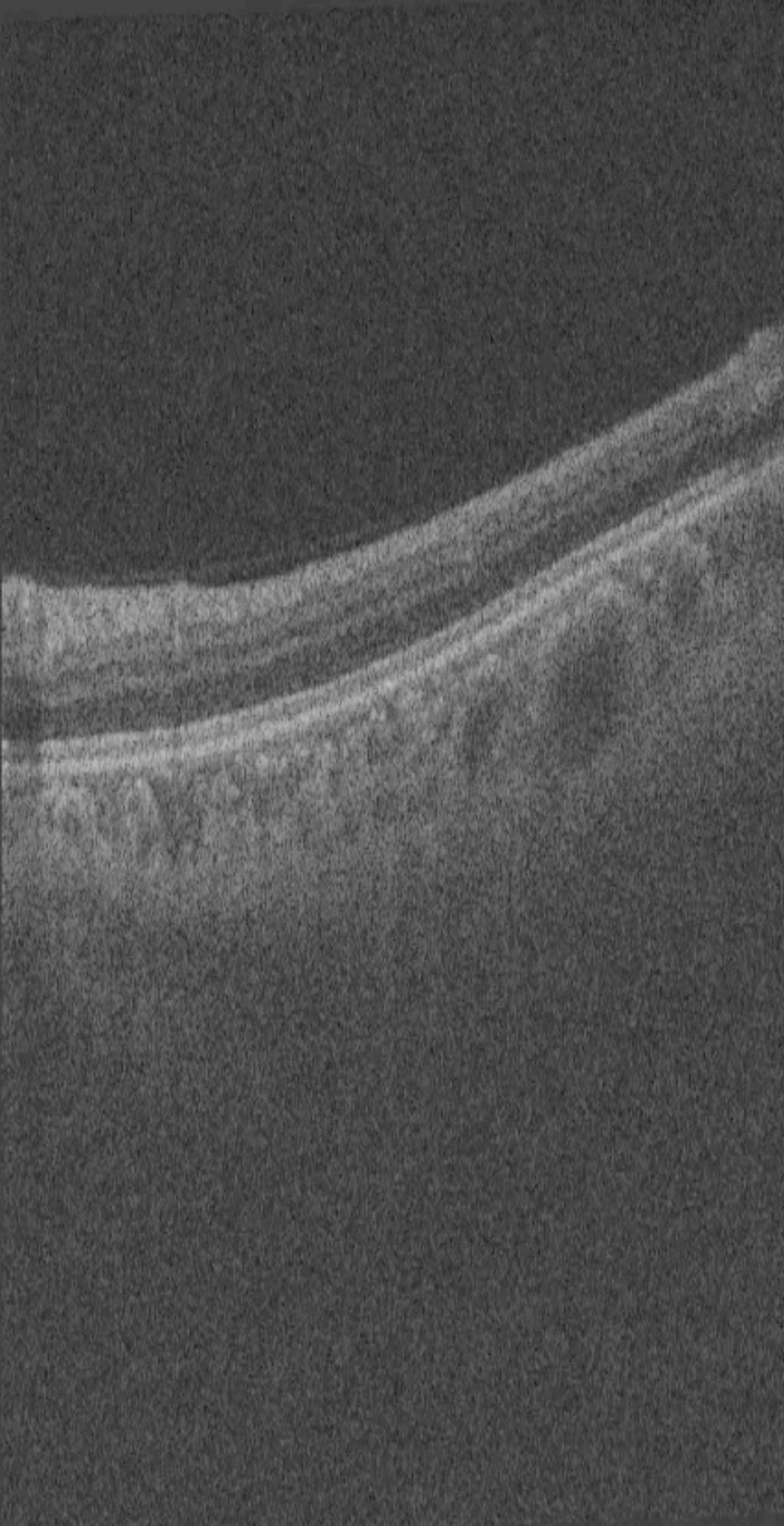


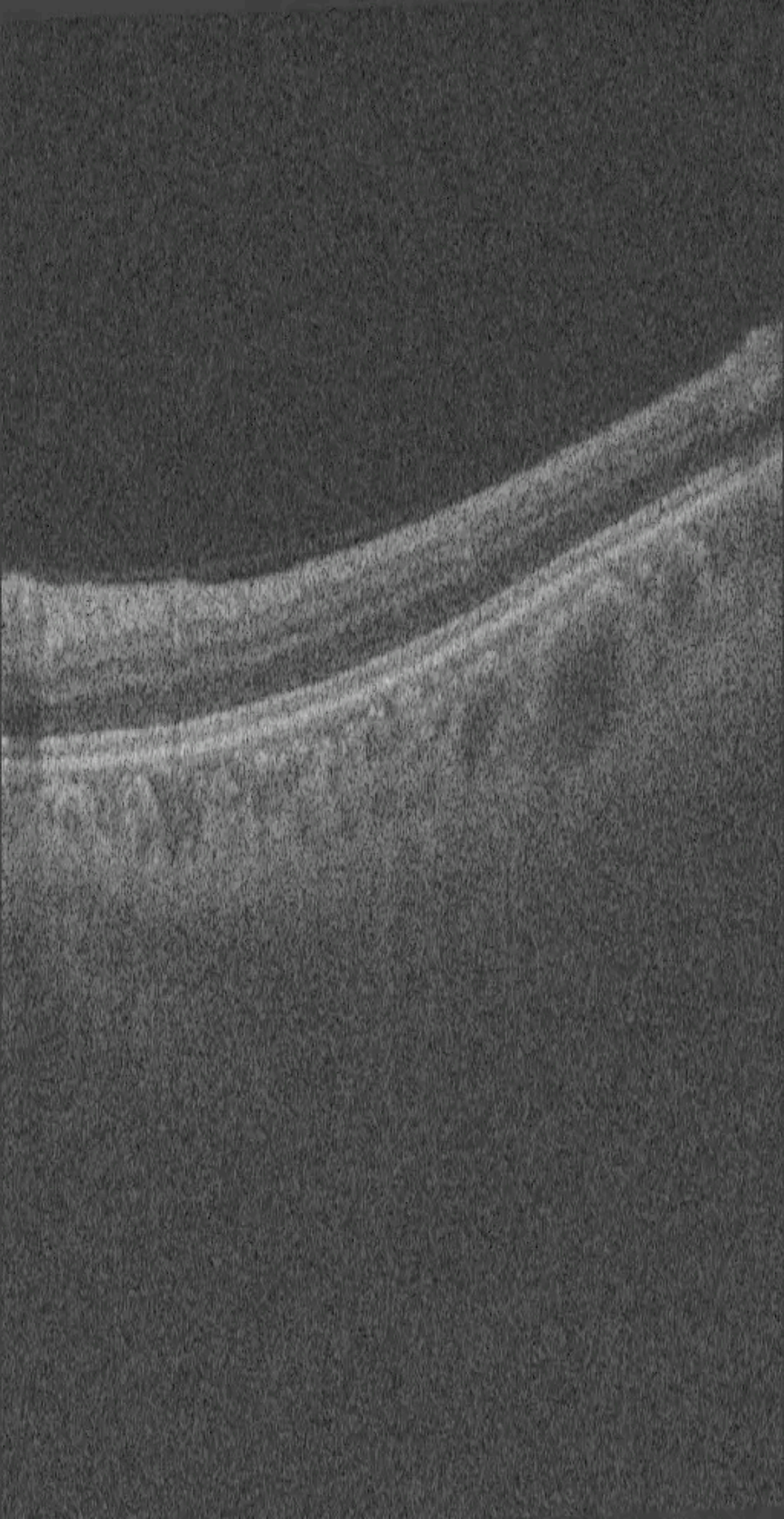
CASEY EYE
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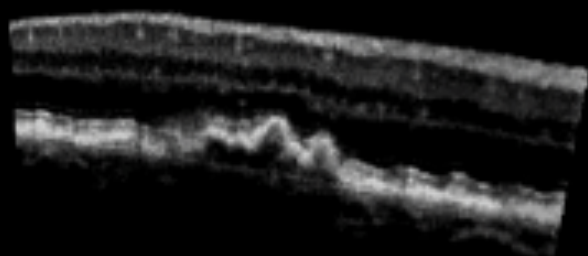
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[Drexler et al., 1999]

PRIMA device in an OCT scan





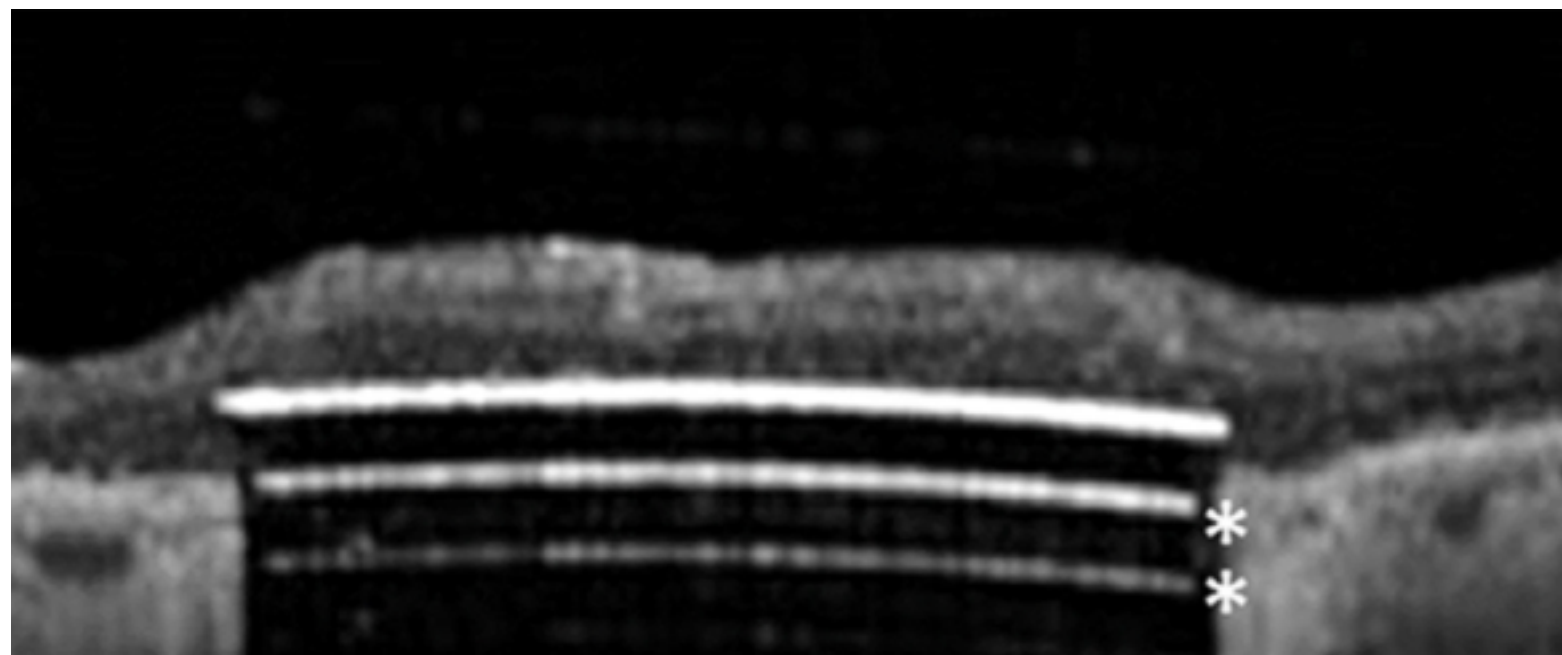


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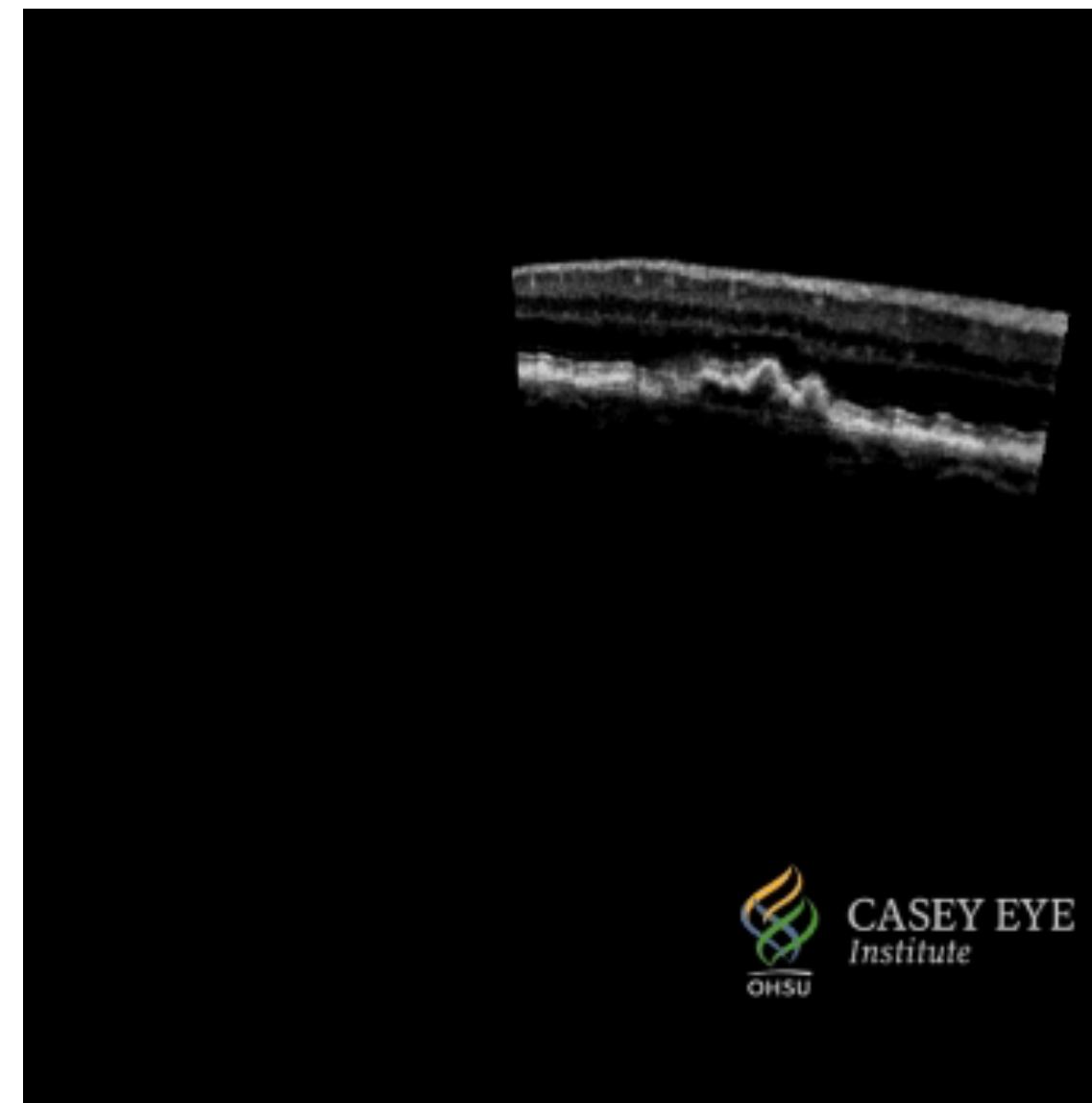
Optical Coherence Tomography (OCT)

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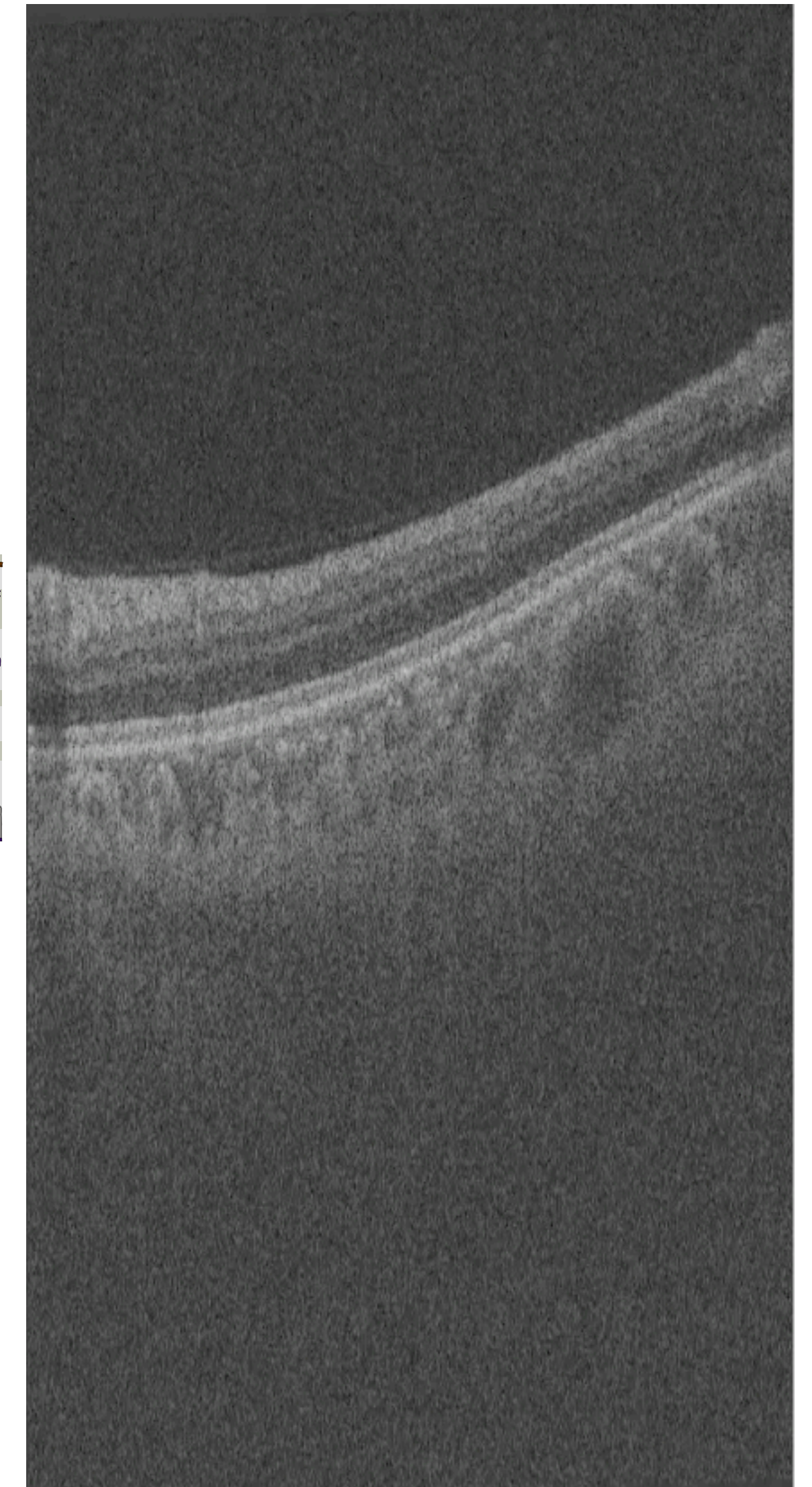
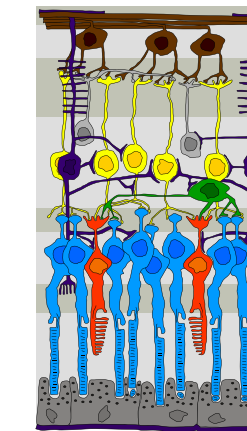
PRIMA device in an OCT scan



[Muqit et al., 2024]



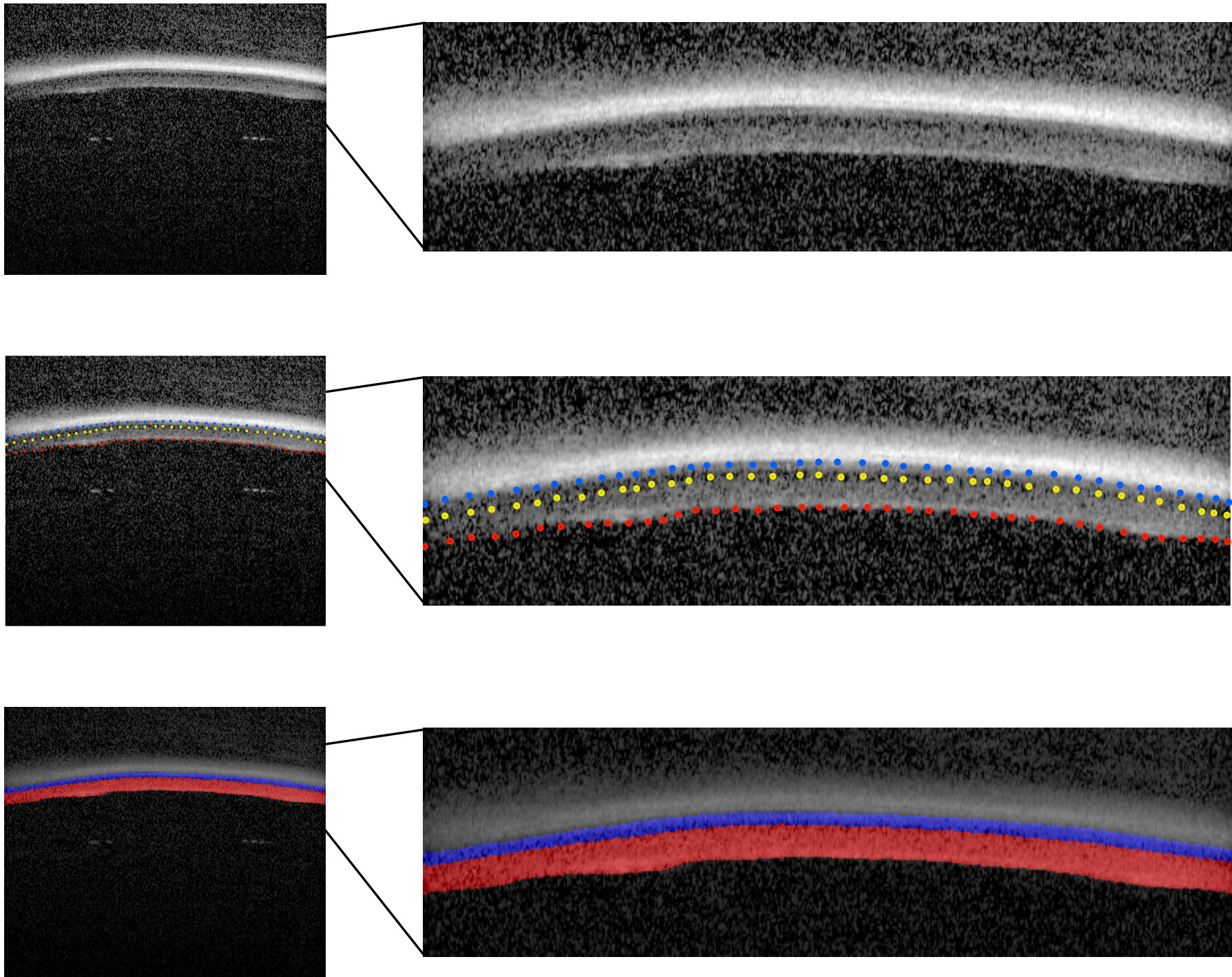
NextOptics, CC BY-SA 4.0 via Wikimedia Commons



[Fang, et al., 2023]

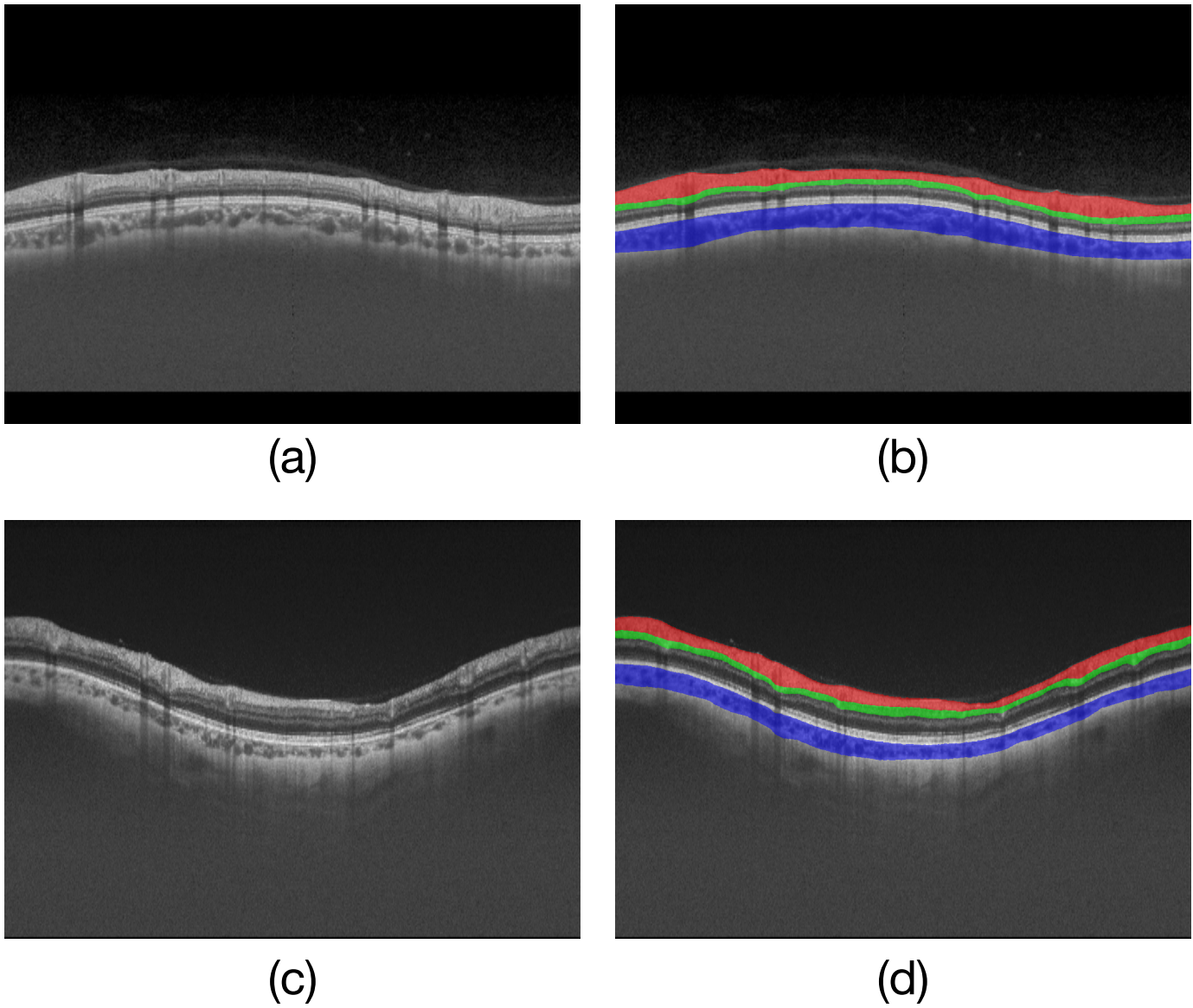
OCT Analysis of *rd10* Mice

Retinal degeneration 10 (rd10) mice:
A model of autosomal recessive
Retinitis Pigmentosa

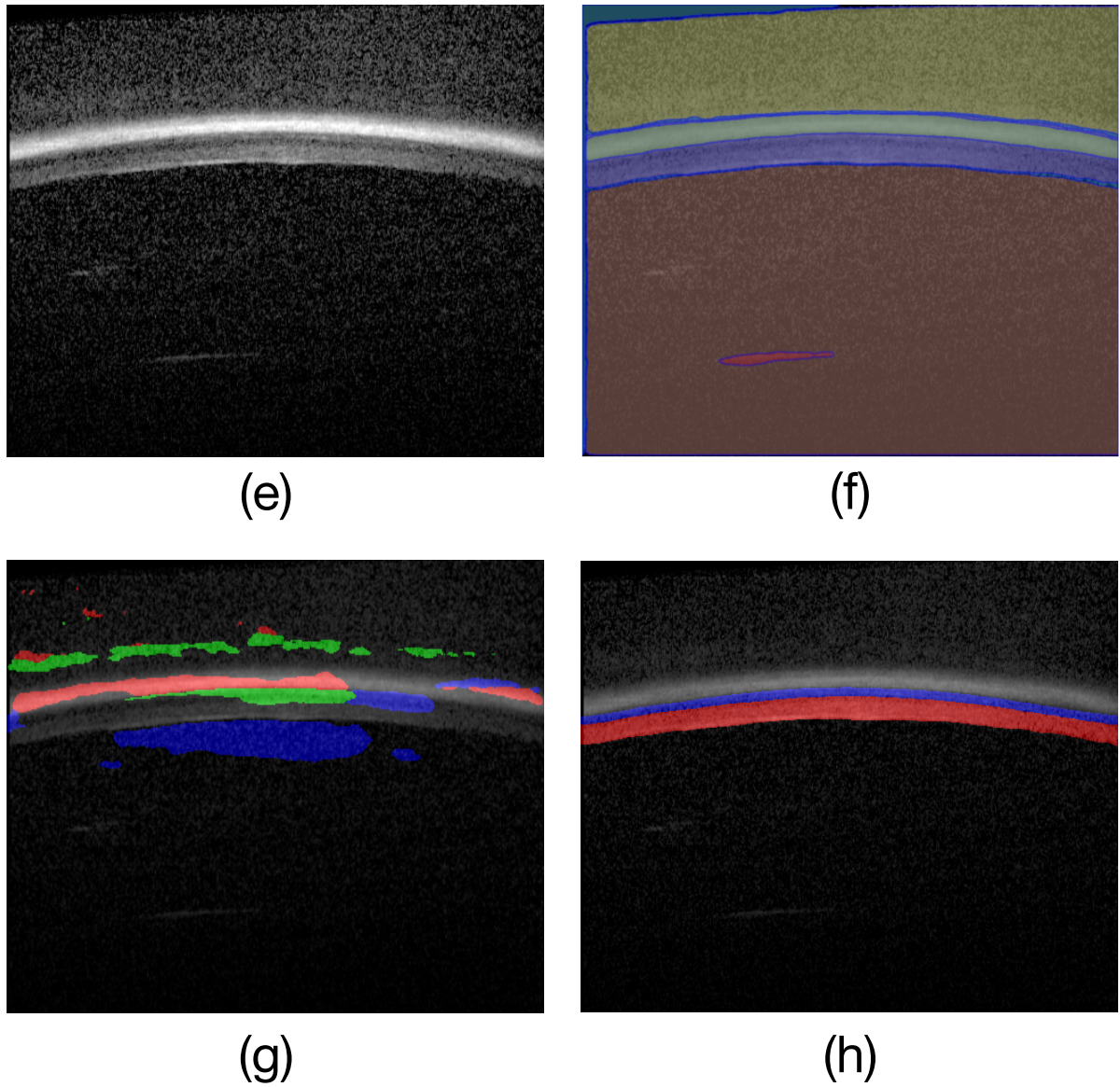


[unpublished]

Human OCT dataset with annotations



rd10 OCT dataset without annotations



Human OCT dataset without annotations

